

FLORALA, AL, USA

WMO: 720361

Lat: 31.043N

Lon: 86.312W

Elev: 314

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.2	30.6	11.5	10.0	35.8	16.7	12.9	37.7	19.9	71.2	17.1	65.8	3.2	350	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	96.5	76.2	94.4	76.1	92.5	75.8	80.5	88.9	79.3	88.4	78.3	87.2	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	149.4	84.1	76.9	141.8	83.1	75.9	136.7	82.3	44.4	88.7	42.9	88.3	41.9	87.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	13.1	11.5	21.1	98.7	3.8	2.9	18.4	100.7	16.2	102.4	14.0	104.1	11.2	106.2		
			19.1	81.5	4.1	2.1	16.1	83.0	13.8	84.2	11.5	85.4	8.5	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.9	50.5	54.9	61.1	67.2	74.6	80.9	82.0	82.0	78.7	68.7	58.4	54.7
	DBStd	13.21	10.42	9.24	8.35	6.20	5.18	3.46	2.75	3.21	4.20	7.50	8.05	9.43
	HDD50	269	124	53	13	1	0	0	0	0	0	1	18	59
	HDD65	1601	461	296	174	48	6	0	0	0	50	229	337	
	CDD50	6792	138	192	358	516	763	926	992	991	862	579	269	206
	CDD65	2649	11	14	53	113	304	476	527	526	412	164	29	20
	CDH74	24527	23	62	388	1011	2765	4746	5196	5128	3573	1368	210	57
	CDH80	10491	0	2	70	231	1121	2301	2398	2350	1567	417	33	1
Wind	WSAvg	4.2	4.7	5.0	5.2	5.1	4.2	3.5	3.1	3.1	3.5	3.9	4.1	4.6
Precipitation	PrecAvg	57.8	5.3	4.9	5.9	4.7	3.7	5.0	6.1	5.1	4.5	3.2	4.3	5.1
	PrecMax	75.8	11.6	11.7	14.2	10.5	9.2	10.3	16.4	10.1	20.6	12.8	11.9	13.5
	PrecMin	40.3	1.6	1.7	0.6	1.8	0.4	1.8	2.0	1.5	1.0	0.2	1.2	1.8
	PrecStd	8.4	2.1	2.2	3.3	2.2	2.1	2.4	3.0	2.0	3.7	2.7	2.5	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	79.8	85.5	88.1	94.9	99.7	97.6	98.9	96.7	91.5	83.9	78.6
		MCWB	66.5	66.6	66.7	68.9	72.2	74.6	76.9	76.9	73.7	73.0	69.4	69.4
	2%	DB	73.8	76.0	81.8	85.0	91.4	96.4	94.9	95.8	93.4	87.6	79.3	75.5
		MCWB	66.3	65.5	68.2	68.4	72.2	76.1	76.9	77.8	74.4	71.7	67.7	67.8
	5%	DB	70.8	72.8	78.4	82.7	88.8	93.5	92.9	93.3	91.0	84.9	75.9	73.0
		MCWB	64.9	63.9	65.8	67.0	71.5	75.7	76.7	77.3	74.4	70.2	66.0	67.2
	10%	DB	67.4	69.9	75.0	80.3	86.4	91.2	90.9	91.0	88.7	81.7	72.9	69.9
		MCWB	62.4	63.0	63.5	65.6	70.8	75.3	76.4	76.8	74.2	68.6	64.0	65.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	69.6	77.9	74.5	79.5	83.7	84.7	82.2	79.0	77.7	73.4	72.6
		MCDB	72.8	74.5	80.8	80.6	83.8	90.4	88.2	92.1	85.9	83.4	78.5	75.2
	2%	WB	68.3	68.4	70.1	72.2	75.1	78.9	80.0	80.4	77.9	75.7	71.4	70.2
		MCDB	72.0	73.1	75.5	79.1	84.2	89.4	89.1	89.8	86.2	81.5	75.7	73.6
	5%	WB	66.1	66.5	68.3	70.8	74.0	77.8	78.7	79.4	77.0	73.9	68.5	68.1
		MCDB	69.9	71.3	74.0	77.5	83.0	88.1	88.3	88.4	85.5	79.8	73.8	71.4
	10%	WB	63.1	64.1	66.2	69.0	73.0	76.8	77.8	78.4	76.1	72.0	66.0	66.0
		MCDB	67.3	68.8	72.7	75.8	82.0	86.7	86.9	87.0	84.2	78.5	71.5	69.4
Mean Daily Temperature Range	5% DB	MDBR	21.0	21.9	22.9	23.0	22.1	20.0	18.5	18.1	19.3	22.4	24.0	19.7
		MCDBR	20.4	22.2	23.9	25.0	24.8	23.3	20.8	21.1	21.8	23.0	23.7	19.7
		MCWBR	13.6	13.9	12.2	10.9	9.0	7.6	6.9	7.1	7.4	9.8	13.9	13.3
	5% WB	MCDBR	17.3	18.7	20.3	19.2	20.1	20.3	18.4	18.4	17.7	18.0	19.6	17.6
	MCWBR	13.6	14.1	14.2	10.9	9.2	8.1	7.5	7.2	7.0	9.6	14.1	14.2	
Clear-Sky Solar Irradiance	taub		0.325	0.336	0.359	0.384	0.431	0.465	0.515	0.499	0.432	0.373	0.346	0.331
	taud		2.532	2.505	2.435	2.371	2.271	2.235	2.107	2.151	2.313	2.454	2.473	2.526
	Ebn at Noon		287	294	293	288	273	263	249	252	266	277	276	279
	Edh at Noon		27	30	35	39	43	45	51	48	39	32	29	26
All-Sky Solar Radiation	RadAvg		956	1142	1541	1874	2058	1960	1879	1747	1602	1404	1081	834
	RadStd		65	146	126	153	130	160	115	120	110	150	94	87

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+180

Nomenclature: See separate page